

Designing a Comprehensive AI Psychologist System

Building a **convincing AI psychologist** goes far beyond simply fine-tuning a small LLM. Research shows that naively fine-tuning a model on limited data often fails to capture the nuanced skills of a therapist ¹. Instead, we must architect a full *psychologist persona* with deep knowledge and techniques. This involves carefully crafting the AI's voice and style, embedding therapy best practices, structuring dialogue for counseling, and ensuring ethical safeguards. We can draw on recent LLM-based counseling systems to guide us.

Crafting a Therapist Persona and Conversational Style

The AI should sound like a warm, empathic therapist, not a generic chatbot. In practice, this means using a **rich system prompt or persona** and conversation rules. For example, one study built two versions of a GPT-based counselor: the second ("Laura") was given a detailed instruction prompt – *"You are a world-class psychologist who is compassionate and understanding. Give answers that confirm the user's feelings and acknowledge their problems... mirror the user's feelings and make them feel you are taking them seriously."* – and this yielded much more supportive, empathetic replies ². Key persona elements include active listening, validation of feelings, patience, and a calm, supportive tone.

- **Empathy and validation:** Always acknowledge the user's emotions and concerns. Echo or paraphrase what they say, as "Laura" did, to build trust ².
- **Reflective listening:** Use phrases like "It sounds like you're feeling..." or "I can hear that this is really tough..." to show understanding.
- **Gentle guidance:** Frame suggestions positively and collaboratively (e.g. "Let's think together..."), rather than dictating.
- **Pacing:** Mirror a real therapist's pacing – ask open questions and wait for input rather than rushing to conclusions ³. For example, in one system ("LLM4CBT"), the AI was explicitly instructed *not* to give immediate solutions but to ask thoughtful, therapy-aligned questions (a key CBT strategy) ³.

By encoding these behaviors in the **system prompt or prompt templates**, we steer the model's tone. For instance, using prompts that explicitly list supportive behaviors (as in Laura's instructions) has been shown to elicit more *empathetic, supportive responses* ². Following proven prompt patterns (e.g. White et al.'s persona pattern) can further reinforce the psychologist role ⁴.

Embedding Therapy Knowledge and Techniques

To truly **simulate a psychologist**, the AI must apply actual therapeutic techniques. We should teach it core methods from different modalities. Important skills include:

- **Cognitive Behavioral Techniques:** Help the user identify and reframe negative thoughts. For example, teach the AI concepts like *"downward arrow"* to drill into automatic thoughts ⁵.

Instruction-based systems have shown that explicitly defining CBT techniques in the prompt (e.g. “when you notice a negative thought, encourage the user to examine evidence for and against it”) guides the model to ask Socratic questions rather than just giving answers ⁵.

- **Active Listening and Reflection:** Emulate therapists by paraphrasing and reflecting the user’s words, which encourages openness. (Laura’s prompt explicitly said to mirror feelings ².)
- **Open-ended Questioning:** Encourage the user to explore their own feelings (similar to motivational interviewing). The LLM4CBT study found that steering the AI to *ask questions* instead of immediately solving problems made it behave more like a human therapist ³.
- **Validation and Normalization:** Let the user know their feelings make sense (e.g. “Many people feel similarly in your situation”). The “Laura” chatbot was instructed to *acknowledge problems and affirm feelings* ².
- **Psychoeducation and Coping Skills:** Include guidance on stress management, relaxation, or mindfulness exercises when appropriate (e.g. encourage deep breathing or suggest a brief mindfulness break). This matches real therapy where psychoeducation is often given.
- **Resource Sharing:** An advanced design might even fetch or suggest external resources. One prototype used a second LLM agent (GPT-4) to determine if a YouTube video or exercise should be offered, then automatically found a relevant link or exercise ⁶. Integrating such tools can make the AI more helpful.

By enumerating these techniques in prompts or training examples, the model learns the *behaviors* of therapy. For example, LLM4CBT’s prompt explicitly included examples of a downward-arrow CBT technique ⁵. We should similarly define rules or examples for the most common therapies (CBT, DBT, motivational interviewing, etc.) so the AI can mimic those strategies.

Training Data and Fine-Tuning Strategy

High-quality, domain-relevant data is crucial. Therapy dialogues are private, so existing datasets are sparse. To create rich training data, one can **augment and synthesize** conversations. For instance, researchers have used GPT-4 with carefully designed prompts to **rewrite real-life text into therapy dialogues**. In one system, social media posts describing personal struggles were converted into multi-turn counseling conversations by prompting GPT-4 to analyze emotions step-by-step and rephrase the content as a dialogue between a user and an AI counselor ⁷. This generated large-scale “psychological counseling dialogue” data that combined authentic experiences with therapist-like responses. We can similarly use creative prompting to expand our dataset: start with anonymous first-person narratives or journals, then instruct an LLM to turn them into a therapist-patient chat.

After building a sizable synthetic dataset, we fine-tune the model on it (potentially using PEFT techniques like LoRA/QLoRA to save resources). One study combining real and GPT-generated dialogues achieved a *comprehensive multi-turn counseling dataset* for instruction fine-tuning ⁸. We should also consider **Reinforcement Learning from Human Feedback (RLHF)**: if we have access to psychologists, they can rate model responses and we can use that feedback to refine the LLM. However, be aware that limited data or resources can make gains hard to see ¹ – the KTH thesis found no significant improvement from RLHF on small data. Thus, gather as much conversation data as possible, and iteratively train and evaluate.

Another key strategy is **knowledge retrieval**. Pure generation risks hallucination, especially in medical contexts. For tasks like diagnosis or factual advice, we can incorporate external knowledge via Retrieval-Augmented Generation (RAG). For example, one system used RAG to fetch relevant parts of the DSM-V

diagnostic criteria based on the user's input, then applied chain-of-thought prompting to produce a structured diagnostic report ⁹. This grounded the AI's output in real psychiatric standards. We can similarly store a knowledge base of counseling guidelines, mental health resources, or anonymized best-practice notes, and retrieve relevant snippets to help the LLM stay accurate.

Chain-of-thought (CoT) prompting is also helpful for reasoning tasks. Prompting the model to “*explain your reasoning*” when answering sensitive questions can mitigate rash replies and align with therapeutic deliberation. InterMind (a depression-assessment system) found that combining RAG with CoT reduced hallucinations and produced more coherent analysis ¹⁰ ⁹. In practice, we could instruct the AI to “think about your suggestions carefully” or break down reasoning in multiple steps when guiding the user.

Maintaining Context and Memory

A real therapist remembers details from past sessions. Likewise, our system should retain **long-term memory of the user**. We can implement a memory module (e.g. a vector database or on-device store) that logs important facts: the user's background, name, ongoing issues, coping strategies tried, past action plans, etc. On each turn, relevant memory entries should be retrieved and fed into the model so it can build on previous progress. Studies of “generative agents” show that LLMs paired with a memory mechanism can produce *consistent, personalized behavior* ¹¹.

For example, in a virtual patient chatbot, memory retrieval helped ensure responses were plausible and aligned with the patient's history ¹¹. In our case, retrieving recent memories (e.g. “last time we talked, you mentioned anxiety at work”) can make the AI's replies feel continuous and attentive. We could use prioritized memory (e.g. important or emotional events get saved) and chunk dialogs as context. On a Raspberry Pi, memory can be a lightweight local database. Using tools like LangChain's memory modules or custom code, the system can maintain a personalized profile. This continuity dramatically deepens the illusion of a real therapeutic relationship.

Ethical Safeguards and Human Oversight

It's crucial to **respect ethical limits**. Every claim should be clearly hypothetical – the AI is a tool, not a licensed counselor. Experts strongly advise that AI chatbots *supplement*, not replace, human therapists ¹² ¹. Research highlights that while chatbots can simulate coping strategies, they *lack true empathy and the ability to read subtle human cues* ¹². We should build in disclaimers (“I'm an AI, not a doctor”) and encourage users to seek professional help for severe issues.

Regular evaluation by mental health professionals is recommended. In one study, licensed psychologists rated AI responses for empathy and appropriateness; surprisingly, a fine-tuned model did *no better* than the base model with a good prompt ¹. This underscores the need for continuous expert feedback. Set up a process where psychologists can review and score the AI's responses on criteria like empathy, accuracy, and usefulness. Use their feedback to refine prompts or retrain components.

Additionally, implement safety triggers. If a user expresses thoughts of self-harm or harm to others, the system should either provide emergency resources or defer to a human. Log interactions securely and handle personal data with privacy in mind (even on-device, be mindful of storage). Recognize the AI's role as

an *assistant*; explicitly instruct it to hand off risk cases to humans. This aligns with best practices in digital mental health.

Deployment and Optimization on Raspberry Pi

A final-year project focusing on on-device deployment must address resource limits. Running an LLM on a Raspberry Pi requires aggressive optimization:

- **Model Choice:** Use a small, efficient model. Research shows that even *lightweight models* (e.g. BART-base, T5-small) can perform well when fine-tuned on counseling tasks ¹³. In that study, a BART-base model generated especially coherent, empathetic responses ¹⁴. Consider a compact model (1-2B parameters) that still captures nuance.
- **Quantization:** Convert the model to low-bit precision. One recent paper (“LLMPi”) demonstrated that quantizing an LLM to 4-bit or even 2-bit weights on a Pi significantly lowered energy use while preserving answer quality ¹⁵ ¹⁶. Applying Post-Training Quantization or Quantization-Aware Training can shrink the model to run within the Pi’s memory and compute constraints.
- **Memory and Storage:** Since space is tight, store any knowledge or memory in compressed form. Use small embeddings, prune unneeded tokens, and consider offloading very large knowledge (like full DSM-V) to a companion server if absolutely necessary.
- **Pipeline Efficiency:** Only run the LLM when needed. For example, pre-load common responses or have a quick rule-based front-end for greetings or simple check-ins. If you implement resource lookup (videos, links), a separate lightweight agent could prepare those offline or on a server.
- **Performance vs Accuracy Trade-offs:** Test different bit-widths or distillation methods to find a sweet spot. The LLMPi results suggest that you can often drop to 4-bit with negligible loss in answer quality ¹⁶. Also, use an efficient inference engine (like ONNX Runtime, TFLite, or dedicated optimized frameworks) tailored for ARM chips.

By combining a quantized model with smart software engineering, you can create a responsive conversational agent on the Pi. The objective is to maintain a smooth chat experience – avoid long pauses – even if it means slightly limiting model size. The cited studies show this is feasible: quantized LLMs on edge hardware can run real-time chat while cutting power use dramatically ¹⁵ ¹⁶.

Conclusion

To maximize the impact of your project, integrate **all** these elements. Design a friendly, human-like persona (using prompting and instruction) that applies proven therapy techniques (CBT questioning, active listening, etc.) in every response ² ⁵. Build a rich training set (via augmented dialogues, real transcripts, and RAG-assisted knowledge) and fine-tune or RLHF accordingly ¹⁰ ⁹. Implement conversation memory so the AI tracks the user’s journey ¹¹. Finally, rigorously test and optimize the model for the Pi’s constraints ¹⁵ ¹⁶.

This comprehensive approach — persona engineering, technique integration, data augmentation, memory modules, and edge optimization — will turn your LLM module into a true *AI psychologist* assistant rather than a toy chatbot. With careful design and iteration, you can convincingly simulate many aspects of a therapist’s role while remaining clear about the AI’s limitations ¹ ¹². In doing so, your FYP component will be both highly complex and meaningful, demonstrating advanced expertise in AI, psychology, and system integration.

Sources: We draw on recent LLM mental-health research and systems (e.g. Woebot-like studies, LLM4CBT, InterMind) to guide this design. Key findings include the importance of persona prompts ², specialized instruction for CBT ⁵, RAG+CoT for accuracy ¹⁰ ⁹, memory for consistency ¹¹, and edge deployment via quantization ¹⁵ ¹⁶. The cited studies illuminate both possibilities and pitfalls in creating AI therapy agents.

¹ diva-portal.org

<https://www.diva-portal.org/smash/get/diva2:1782678/FULLTEXT01.pdf>

² ⁴ ⁶ diva-portal.org

<https://www.diva-portal.org/smash/get/diva2:1789084/FULLTEXT01.pdf>

³ ⁵ Frontiers | Aligning large language models for cognitive behavioral therapy: a proof-of-concept study

<https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2025.1583739/full>

⁷ ⁸ ⁹ ¹⁰ InterMind: A Doctor-Patient-Family Interactive Depression Assessment System Empowered by Large Language Models

<https://arxiv.org/html/2409.14878v1>

¹¹ Integrating large language model-based agents into a virtual patient chatbot for clinical anamnesis training - PubMed

<https://pubmed.ncbi.nlm.nih.gov/40547455/>

¹² Can AI chatbots replace human therapists? Unpacking the mental health revolution

<https://sundaytribune.co.za/lifestyle/health/2025-07-09-can-ai-chatbots-replace-human-therapists-unpacking-the-mental-health-revolution/>

¹³ ¹⁴ Exploring the potential of lightweight large language models for AI-based mental health counselling task: a novel comparative study | Scientific Reports

https://www.nature.com/articles/s41598-025-05012-1?error=cookies_not_supported&code=a637ecb8-934a-44c1-af10-3a57004a2be2

¹⁵ ¹⁶ LLMPi: Optimizing LLMs for High-Throughput on Raspberry Pi

<https://arxiv.org/html/2504.02118v1>